

Qualitative data

Qualitative data describe categories.

Why is a course subject a category even if the school assigns it a number?

Quantitative data

Quantitative data record numerical amounts.

Does the number 201 mean the same kind of thing in “room 201” and “201 seats”?

Discrete data

Discrete quantitative values belong to a separated, countable set.

Can the number of open browser tabs be 4.5?

Continuous data

Continuous quantities can take any value within an interval.

A download lasts 3.2 seconds. Could a more precise timer report 3.24 seconds?

Classifying a variable

A spreadsheet records each classroom's building name.

What type of data is this?

Classifying a count

A spreadsheet records the number of chairs in each classroom.

What type of data is this?

Classifying a measurement

A spreadsheet records each classroom's floor area in square meters.

What type of data is this?

Numbers as labels

Students have identification numbers such as 10482 and 10517.

Would the mean ID number describe a meaningful amount?

Bar graphs

A bar graph compares category counts or percentages using bar lengths.

For 12 blue, 8 black, and 5 red notebooks, what would the bars show?

Pie charts

A pie chart represents mutually exclusive parts of one whole.

Among 25 notebooks, 12 are blue, 8 black, and 5 red. What fraction belongs to each slice?

Pareto charts

A Pareto chart orders category bars from largest to smallest.

How would you arrange categories with counts 7, 19, 4, and 11?

Overlapping categories

Among 100 students, 65 use laptops and 55 use tablets. Some use both.

Would these percentages make a valid two-slice pie chart?

Missing responses

A class has 30 students. A device survey records 18 laptops and 9 tablets, with one response per student.

What is missing from this summary?

Simple random sampling

Every possible sample of the chosen size has the same chance of selection.

How could we choose 5 of 30 numbered students this way?

Random number selection

Choose 4 different IDs from 01 through 20.
Generated IDs: 07, 07, 23, 14, 02, 19.

Which IDs enter the sample?

Stratified sampling

Randomly sample individuals within every stratum.

How could we sample 10% of students from each academic program?

Cluster sampling

Randomly select clusters and include everyone in those clusters.

What sample results if we randomly choose three class sections and survey everyone in them?

Systematic sampling

After a random start, select every k th member of a list.

On a list of 60 students, start at 4 and select every tenth student. Which IDs enter the sample?

Convenience sampling

A convenience sample uses readily available individuals.

What could go wrong if a campus survey uses only students in the nearest classroom?

Voluntary response

A voluntary-response sample consists of people who choose to respond to an open invitation.

Who might respond to an online poll about parking fees?

Strata or clusters?

Plan A samples 5 students from every program. Plan B randomly chooses 2 programs and surveys everyone in them.

Which plan uses clusters?

Sampling variation

Different random samples can produce different statistics.

Two random samples report 42% and 47% support for a proposal. Must one be wrong?

Variation in data

Observations within one sample can differ.

Three printers take 9, 12, and 18 seconds for the same file. What might account for the variation?

Bias

Bias is a systematic tendency to misrepresent the target population or quantity.

Would surveying 10,000 library visitors remove bias in a survey of all students?

Nonresponse

A college randomly invites 400 students to a survey. Only 35 reply.

What concern remains even though the invitations were random?

Question wording

“Should the college waste money on longer library hours?”

How could we ask about the same issue more neutrally?

A sampling choice

A college wants feedback from both online and on-campus students.

What sampling plan would you propose?